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Machine Learning

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The slides of this chapter
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Springer

Chapter 12

Computational

Learning Theory

Outline

□ Overview

- Concerns
- Some concepts and notations

□ Learnability

- What is learning
- What is learnable
- The impact of the complexity of hypothesis spaces on learnability
 - Finite hypothesis space
 - Infinite hypothesis space: VC dimension
 - Infinite hypothesis space: Rademacher complexity

□ Stability

Concerns

- ❑ How to describe the process of **learning**?
- ❑ What kind of problems are **learnable**?
- ❑ What kind of problem is **easy to learn**?
- ❑ For a given learning algorithm, can its performance be predicted theoretically?
- ❑ How can the theoretical results guide the algorithm design of practical problems?

Some concepts and notations

- Sample set: independent and identically distributed samples, only considering the binary classification problem

$$D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}, \mathbf{x}_i \in \mathcal{X}, y_i \in \mathcal{Y} = \{-1, +1\}.$$

- h is a mapping from $\mathcal{X}$ to $\mathcal{Y}$

- **Generalization error:** expected error of classifier

$$E(h; \mathbf{D}) = P_{\mathbf{x} \sim \mathbf{D}}(h(\mathbf{x}) \neq y)$$

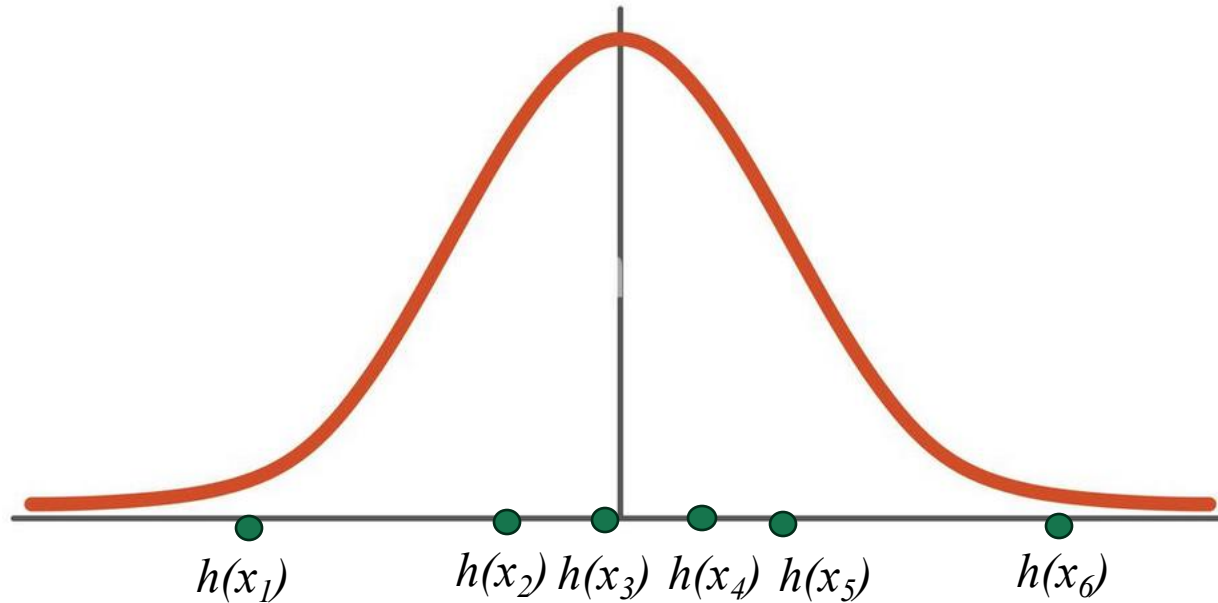
- **Empirical error:** the average error of the classifier on a given sample set

$$\hat{E}(h; \mathcal{D}) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(h(\mathbf{x}_i) \neq y_i)$$

Since D is independent and identically distributed sampling from $\mathbf{D}$, the expectation of empirical error of h is equal to its generalization error.

When writing is not misleading, $E(h; \mathbf{D})$ and $\hat{E}(h; \mathcal{D})$ are abbreviated as $E(h)$ and $\hat{E}(h)$ respectively

Expected error $E(h; \mathcal{D}) = P_{x \sim \mathcal{D}}(h(x) \neq y)$



Some concepts and notations

❑ Error parameter ϵ

ϵ is an upper bound of $E(h)$, where $E(h) \leq \epsilon$

$\Rightarrow$ The maximum error we can tolerate for a learned model

❑ Consistency

A mapping h is said to be consistent with D if the empirical error of h on the data set D is 0.

❑ Disagreement

For any two mappings $h_1, h_2 \in \mathcal{X} \rightarrow \mathcal{Y}$, their difference can be measured by the disagreement

$$d(h_1, h_2) = P_{x \sim \mathcal{D}}(h_1(\mathbf{x}) \neq h_2(\mathbf{x}))$$

What is learning

□ Concept

Concept provides a mapping from the sample space $\mathcal{X}$ to the label space $\mathcal{Y}$, and it determines the ground-truth label y of the sample x .

● Target concept

A concept c is said to be a target concept if $c(x) = y$ holds for every sample (x, y) .

● Concept class

The set of all target concepts that we wish to learn is called a concept class, denoted by $\mathcal{C}$.

What is learning

❑ Concept

Concept provides a mapping from the sample space $\mathcal{X}$ to the label space $\mathcal{Y}$, and it determines the ground-truth label y of the sample x .

❑ For the Watermelon example:

ID	Color	Root	Sound	Ripe (Y)
1	green	curly	muffled	true
2	dark	curly	muffled	true
3	green	straight	crisp	false
4	dark	slightly curly	dull	false

- ❑ The concept to be learned (ripeness) could be expressed on the three features; “ripe is watermelon with color=A, Root=B, and Sound=C”, and from data we should replace variables A, B, and C with values that do not contradict the evidence of the data

What is learning

□ Hypothesis space

The set of all possible concepts for a given learning algorithm $\mathcal{L}$ is called a hypothesis space, denoted by $\mathcal{H}$.

- Since the ground-truth concept class is unknown to learning algorithms, $\mathcal{H}$ and $\mathcal{C}$ are usually different. A learning algorithm constructs $\mathcal{H}$ by collecting all concepts that are believed to be the target concepts.
- Since it is unknown whether the collected concepts are ground-truth target concepts, $h \in \mathcal{H}$ is referred to as a hypothesis, which provides a mapping from the sample space $\mathcal{X}$ to the label space $\mathcal{Y}$.

What is learning

□ Separable and non-separable

- Separable

If $c \in \mathcal{H}$, then $\mathcal{H}$ contains a hypothesis that can correctly classify all instances, and such a learning problem is said to be *separable* or *consistent* with respect to the learning algorithm $\mathcal{L}$.

- Non-separable

If $c \notin \mathcal{H}$, then $\mathcal{H}$ does not contain any hypothesis that can correctly classify all instances, and such a learning problem is said to be *non-separable* or *inconsistent* with respect to the learning algorithm $\mathcal{L}$.

What is learning

- Given a training set D , we wish the learning algorithm $\mathcal{L}$ can learn a hypothesis h that is close to the target concept c .

Why not learn the exact target concept c ?

The process of machine learning is restricted by many factors

- Since the training set D usually contains finite samples, there often exist many *equivalent hypotheses* that cannot be distinguished by learning algorithms on D ;
- Also, there **exists some randomness when sampling** D from $\mathcal{D}$, and hence the hypotheses learned from different equal-sized training sets could be different.

What is learnable

□ Probably Approximately Correct, PAC

We hope to learn a better model with a greater grasp, that is, a hypothesis that is **probably approximately correct**.

Let δ denote the confidence, and the formal definition as follows:

Definition PAC Identify

A learning algorithm $\mathcal{L}$ is said to PAC identify the concept class $\mathcal{C}$ from the hypothesis space $\mathcal{H}$ if, for any $c \in \mathcal{C}$ and distribution $\mathcal{D}$, and $0 < \epsilon, \delta < 1$, the learning algorithm $\mathcal{L}$ outputs a hypothesis $h \in \mathcal{H}$ satisfying:

$$P(E(h) \leq \epsilon) \geq 1 - \delta,$$

Such a learning algorithm $\mathcal{L}$ has a probability of at least $1 - \delta$ of learning an approximation of the target concept c with an error of at most ϵ .

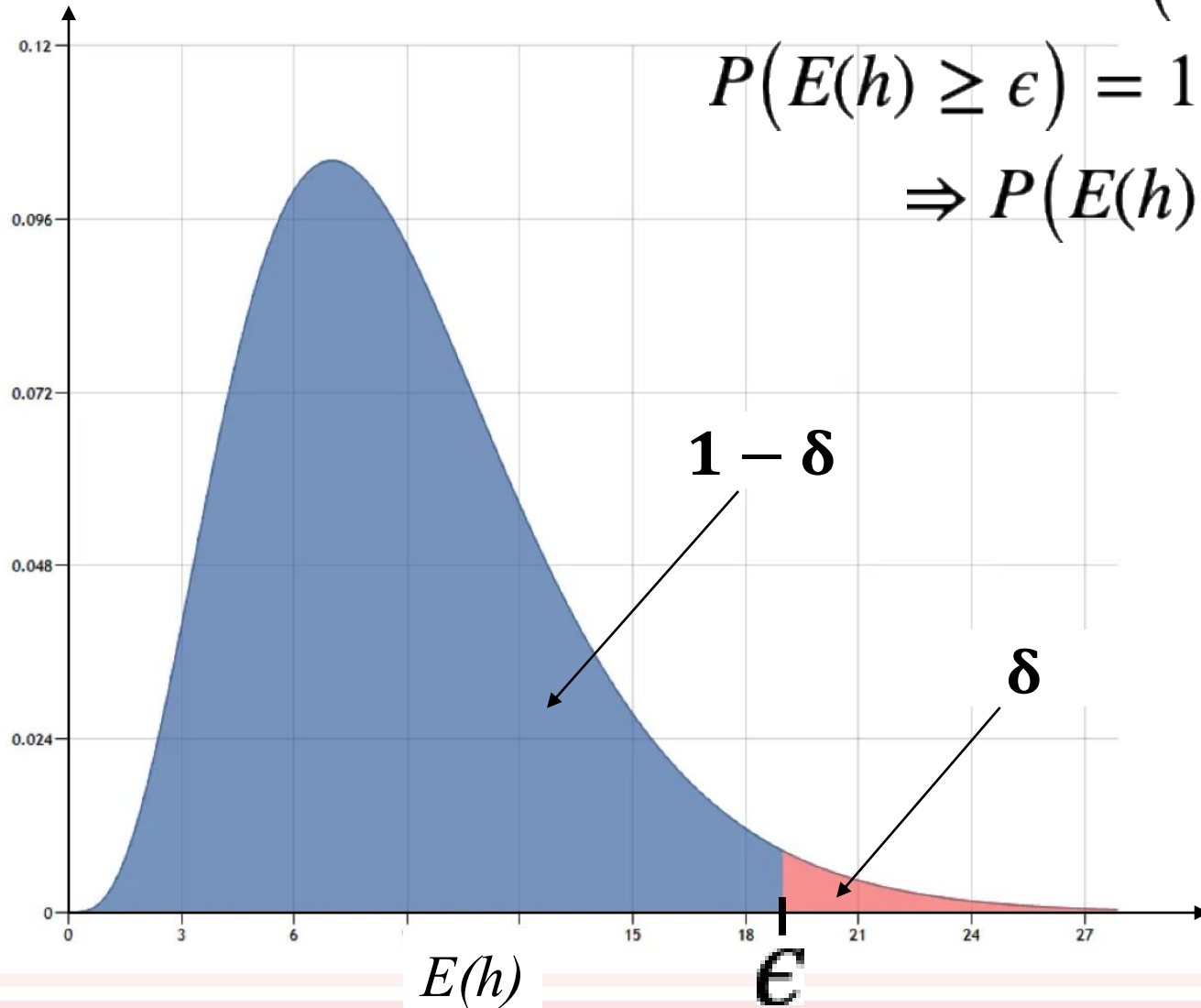
Probability hypothesis h approximately correct

Probability $P(E(h))$

$$P(E(h) \geq \epsilon) \leq \delta$$

$$P(E(h) \geq \epsilon) = 1 - P(E(h) \leq \epsilon)$$

$$\Rightarrow P(E(h) \leq \epsilon) \geq 1 - \delta$$



Relationship among the main dimensions of the problem

- The definition of PAC learnable provides a relationship among the main dimensions of the learning problem:
 - sample size m
 - the error ϵ ,
 - the confidence $1 - \delta$
 - the complexity of data $\text{size}(x)$
 - the complexity of target concept $\text{size}(c)$.

What is learnable

Definition PAC Learnable

A target **concept class** $\mathcal{C}$ is said to be PAC learnable **with respect to the hypothesis space** $\mathcal{H}$ if there **exists** a learning algorithm $\mathcal{L}$ such that:

for any $0 < \epsilon, \delta < 1$ and distribution $\mathcal{D}$, the learning algorithm $\mathcal{L}$ can PAC identify the concept class $\mathcal{C}$ from the hypothesis space $\mathcal{H}$, for any $m \geq \text{poly}(1/\epsilon, 1/\delta, \text{size}(\mathbf{x}), \text{size}(c))$, where $\text{poly}(\cdot, \cdot, \cdot, \cdot)$ is a polynomial function and m is the number of *i.i.d.* training samples drawn from the distribution $\mathcal{D}$.

What is learnable

For learning algorithms, it is necessary to consider the **running time complexity**. Hence, we further define:

Definition PAC Learning Algorithm

A concept class $\mathcal{C}$ is said to be **efficiently PAC learnable** by its PAC learning algorithm $\mathcal{L}$ if $\mathcal{C}$ is PAC learnable by $\mathcal{L}$ within a polynomial time $\text{poly}(1/\epsilon, 1/\delta, \text{size}(\mathbf{x}), \text{size}(c))$.

What is learnable

Suppose the learning algorithm $\mathcal{L}$ processes each sample with a constant time, then the running time complexity is equivalent to the sample complexity, and we could focus only on the sample complexity:

Definition Sample Complexity

The sample complexity of a PAC learning algorithm $\mathcal{L}$ is the smallest sample size $m \geq \text{poly}(1/\epsilon, 1/\delta, \text{size}(\mathbf{x}), \text{size}(c))$ required by $\mathcal{L}$.

What is learnable

□ The significance of PAC learning :

- PAC learning provides a formal framework for describing the learning ability of learning algorithms, and many important questions can be discussed theoretically under this framework.
 - What are the requirements for learning a good model for a given learning problem?
 - What are the conditions for an algorithm to learn effectively?
 - How many training samples are required to learn a good model?
- The analysis of **time complexity** of algorithms is transformed into the analysis of **sample complexity** (see slide 21)

Finite Hypothesis Space

- The complexity of hypothesis space $\mathcal{H}$ is one of the important factors that affect learnability.
 - In general, a larger $\mathcal{H}$ is more likely to contain the target concept we are looking for, though the larger hypothesis space also makes it more difficult to find the target concept.
 - $\mathcal{H}$ is called a finite hypothesis space if $|\mathcal{H}|$ is finite, and an infinite hypothesis space otherwise.

Finite Hypothesis Space

- The complexity of hypothesis space $\mathcal{H}$ is one of the important factors that affect the difficulty of learning task

- Properly PAC learnable

If the hypothesis space is the same as the concept class (i.e $\mathcal{H} = \mathcal{C}$), then $\mathcal{C}$ is said to be properly PAC learnable with respect to $\mathcal{H}$.

- Intuitively, it means the ability of the learning algorithm properly matches the learning problem.
- However, in practical application, we usually know nothing about concept class, so it is impractical to design a learning algorithm with the same assumption space as concept class.

- Therefore, it is more realistic to study the cases when the hypothesis space and the concept class are different $\mathcal{H} \neq \mathcal{C}$.

Finite Hypothesis Space

□ Separable case:

The target concept c is in the hypothesis space $\mathcal{H}$ (i.e. $c \in \mathcal{H}$).

Given a training set D with size m , how can we find a hypothesis satisfying the constraint of a given error parameter?

□ Natural learning strategy

- Since the labels of the samples in D are assigned by the target concept $c \in \mathcal{H}$, any hypotheses that misclassify any samples in D must not be the target concept c .
- Simply eliminate all hypotheses that are inconsistent with D and keep the rest.
- When the training set D is sufficiently large, we can keep eliminating inconsistent hypotheses from $\mathcal{H}$ until there is only one hypothesis left, which must be the target concept c .

Finite Hypothesis Space (1)

- In general, due to the limited size of training set D , there may be more than one equivalent hypothesis in the hypothesis space $\mathcal{H}$, and it is impossible to further distinguish the advantages and disadvantages of these false equivalent hypotheses.

How many samples do we need to learn a good approximation of the target concept c ?

- We first estimate the probability of having a hypothesis that performs perfectly on the training set
- Suppose the generalization error of a hypothesis is higher than ϵ
- For any i.i.d. sample drawn from the distribution D , we have:

$$\begin{aligned} P(h(\mathbf{x}) = y) &= 1 - P(h(\mathbf{x}) \neq y) \\ &= 1 - E(h) \\ &< 1 - \epsilon \end{aligned}$$

Finite Hypothesis Space (2)

- Since D contains m i.i.d samples, the probability that h and D are consistent is given by:

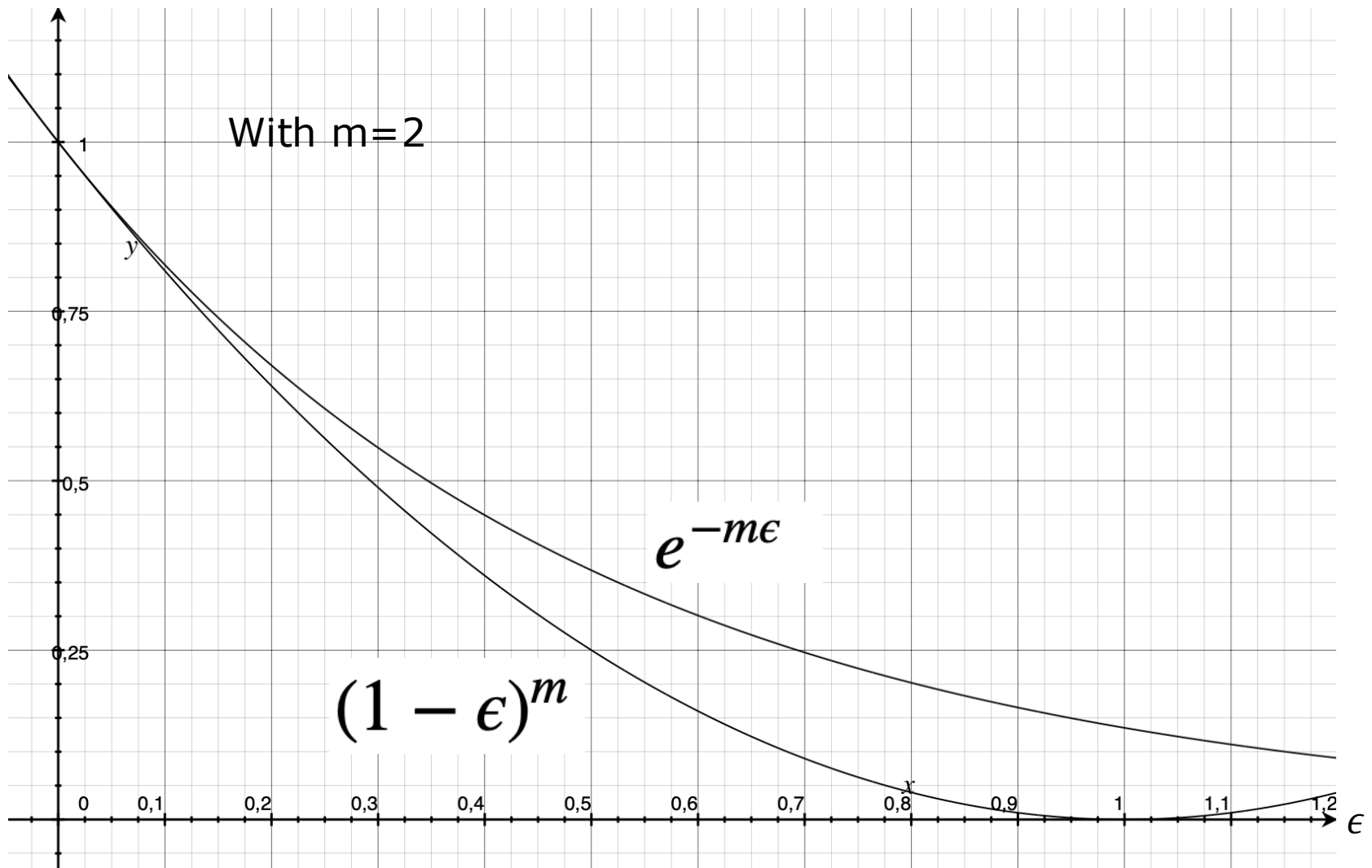
$$\begin{aligned} &P(h(\mathbf{x}_1) = y_1) \wedge P(h(\mathbf{x}_2) = y_2) \wedge \cdots \wedge P(h(\mathbf{x}_m) = y_m) \\ &= [1 - P(h(\mathbf{x}) \neq y)]^m \\ &< (1 - \epsilon)^m \end{aligned}$$

- We need to ensure that the total probability of having any hypotheses that are consistent with D and have generalization error greater than ϵ is not greater than δ .

$$\begin{aligned} P(h \in \mathcal{H} : E(h) > \epsilon \wedge \hat{E}(h) = 0) &< |\mathcal{H}| (1 - \epsilon)^m \\ &< |\mathcal{H}| e^{-m\epsilon} \end{aligned}$$

$$|\mathcal{H}| e^{-m\epsilon} \leq \delta \quad \Rightarrow \quad m \geq \frac{1}{\epsilon} (\ln |\mathcal{H}| + \ln \frac{1}{\delta})$$

$$(1 - \varepsilon)^m \leq e^{(-m\varepsilon)}$$



Finite Hypothesis Space (3)

$$m \geq \frac{1}{\epsilon} \left(\ln |\mathcal{H}| + \ln \frac{1}{\delta} \right).$$

- The above found equation, shows that every finite hypothesis space $\mathcal{H}$ is PAC learnable
- As the number of samples increases, the generalization error of the output hypothesis h converges towards 0 at a convergence rate of $O(\frac{1}{m})$.

Finite Hypothesis Space (1)

□ Non-separable Case

For difficult learning problems, the target concept c is usually not in the hypothesis space $\mathcal{H}$. Suppose $h \in \mathcal{H}, \hat{E}(h) \neq 0$

because
$$\hat{E}(h; \mathcal{D}) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(h(\mathbf{x}_i) \neq y_i)$$

that is, every hypothesis in $\mathcal{H}$ misclassifies at least one training example.

To prove the next property we need to recall

Hoeffding inequality:

Given m i.i.d variables x_i , such that: $0 \leq x_i \leq 1$, for any $\epsilon > 0$,

$$P\left(E(h) - \hat{E}(h) \geq \epsilon\right) \leq \exp(-2m\epsilon^2) \quad P\left(\hat{E}(h) - E(h) \geq \epsilon\right) \leq \exp(-2m\epsilon^2)$$
$$P\left(|E(h) - \hat{E}(h)| \geq \epsilon\right) \leq 2\exp(-2m\epsilon^2)$$

Finite Hypothesis Space (2)

□ Non-separable Case

Applying Hoeffding inequality to $E(h)$:

$$1. P\left(|E(h) - \hat{E}(h)| \geq \epsilon\right) \leq 2\exp(-2m\epsilon^2) \Rightarrow P(E(h) \geq \epsilon) \leq 2\exp(-2m\epsilon^2)$$

$$2. P(E(h) \leq \epsilon) \geq 1 - \delta \quad \Rightarrow P(E(h) \geq \epsilon) = 1 - P(E(h) \leq \epsilon)$$

$$\Rightarrow P(E(h) \geq \epsilon) \leq \delta \quad \Rightarrow \delta = 2\exp(-2m\epsilon^2)$$

$$\Rightarrow \epsilon = \pm \sqrt{\frac{1}{2m} \ln\left(\frac{\delta}{2}\right)}$$

$$P\left(|E(h) - \hat{E}(h)| \geq \epsilon\right) \leq 2\exp(-2m\epsilon^2) \Rightarrow$$
$$\hat{E}(h) - \sqrt{\frac{1}{2m} \ln\left(\frac{\delta}{2}\right)} \leq E(h) \leq \hat{E}(h) + \sqrt{\frac{1}{2m} \ln\left(\frac{\delta}{2}\right)}$$

It shows that, for a large m , the empirical error of $\hat{E}(h)$ is a good approximation to its generalization error $E(h)$.

Finite Hypothesis Space

□ Non-separable Case

Theorem 12.1

Let $\mathcal{H}$ be a finite hypothesis space. Then, for any $h \in \mathcal{H}$ and $0 < \delta < 1$, we have $P\left(|E(h) - \hat{E}(h)| \leq \sqrt{\frac{\ln |\mathcal{H}| + \ln(2/\delta)}{2m}}\right) \geq 1 - \delta$.

Proof:

let $h_1, h_2, \dots, h_{|\mathcal{H}|}$ be the hypothesis in $\mathcal{H}$:

The probability $P\left(\exists h \in \mathcal{H} : |E(h) - \hat{E}(h)| \geq \epsilon\right)$
 $P\left(|E(h_1) - \hat{E}(h_1)| \geq \epsilon\right)$

Finite Hypothesis Space

□ Non-separable Case

Theorem 12.1

Let $\mathcal{H}$ be a finite hypothesis space. Then, for any $h \in \mathcal{H}$ and $0 < \delta < 1$, we have $P\left(|E(h) - \hat{E}(h)| \leq \sqrt{\frac{\ln |\mathcal{H}| + \ln(2/\delta)}{2m}}\right) \geq 1 - \delta$.

Proof:

let $h_1, h_2, \dots, h_{|\mathcal{H}|}$ be the hypothesis in $\mathcal{H}$:

The probability $P\left(\exists h \in \mathcal{H} : |E(h) - \hat{E}(h)| \geq \epsilon\right)$
 $P\left((|E(h_1) - \hat{E}(h_1)| \geq \epsilon) \vee (|E(h_2) - \hat{E}(h_2)| \geq \epsilon)\right)$

Finite Hypothesis Space

□ Non-separable Case

Theorem 12.1

Let $\mathcal{H}$ be a finite hypothesis space. Then, for any $h \in \mathcal{H}$ and $0 < \delta < 1$, we have $P\left(|E(h) - \hat{E}(h)| \leq \sqrt{\frac{\ln |\mathcal{H}| + \ln(2/\delta)}{2m}}\right) \geq 1 - \delta$.

Proof:

let $h_1, h_2, \dots, h_{|\mathcal{H}|}$ be the hypothesis in $\mathcal{H}$:

The probability $P\left(\exists h \in \mathcal{H} : |E(h) - \hat{E}(h)| \geq \epsilon\right)$

$$P\left((|E(h_1) - \hat{E}(h_1)| \geq \epsilon) \vee (|E(h_2) - \hat{E}(h_2)| \geq \epsilon) \vee \dots \vee (|E(h_{|\mathcal{H}|}) - \hat{E}(h_{|\mathcal{H}|})| \geq \epsilon)\right)$$

Finite Hypothesis Space

□ Non-separable Case

Theorem 12.1

Let $\mathcal{H}$ be a finite hypothesis space. Then, for any $h \in \mathcal{H}$ and $0 < \delta < 1$, we have $P\left(|E(h) - \hat{E}(h)| \leq \sqrt{\frac{\ln |\mathcal{H}| + \ln(2/\delta)}{2m}}\right) \geq 1 - \delta$.

Proof:

let $h_1, h_2, \dots, h_{|\mathcal{H}|}$ be the hypothesis in $\mathcal{H}$:

$$\begin{aligned} &\text{The probability } P\left(\exists h \in \mathcal{H} : |E(h) - \hat{E}(h)| \geq \epsilon\right) \\ &P\left((|E(h_1) - \hat{E}(h_1)| \geq \epsilon) \vee (|E(h_2) - \hat{E}(h_2)| \geq \epsilon) \vee \dots \vee (|E(h_{|\mathcal{H}|}) - \hat{E}(h_{|\mathcal{H}|})| \geq \epsilon)\right) \\ &\leq \sum_{h \in \mathcal{H}} P(|E(h) - \hat{E}(h)| \geq \epsilon) \\ &\text{Applying Hoeffding inequality: } P(|E(h) - \hat{E}(h)| \geq \epsilon) \leq 2\exp(-2m\epsilon^2) \\ &\sum_{h \in \mathcal{H}} P(|E(h) - \hat{E}(h)| \geq \epsilon) \leq 2|\mathcal{H}|\exp(-2m\epsilon^2) \end{aligned}$$

Finite Hypothesis Space

□ Non-separable Case

From Hoeffding inequality:

$$\sum_{h \in \mathcal{H}} P(|E(h) - \hat{E}(h)| \geq \epsilon) \leq 2|\mathcal{H}| \exp(-2m\epsilon^2)$$

$$\Rightarrow \sum_{h \in \mathcal{H}} P(|E(h) - \hat{E}(h)| \leq \epsilon) \geq \underbrace{1 - 2|\mathcal{H}| \exp(-2m\epsilon^2)}_{\delta}$$

The theorem is proved by letting $\delta = 2|\mathcal{H}| \exp(-2m\epsilon^2)$

$$\Rightarrow \frac{\delta}{2|\mathcal{H}|} = \exp(-2m\epsilon^2) \Rightarrow \ln\left(\frac{\delta}{2|\mathcal{H}|}\right) = -2m\epsilon^2 \Rightarrow -\ln\left(\frac{\delta}{2|\mathcal{H}|}\right) = 2m\epsilon^2$$

$$\Rightarrow \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) = 2m\epsilon^2 \Rightarrow \ln\left(\frac{2}{\delta}\right) + \ln(|\mathcal{H}|) = 2m\epsilon^2 \Rightarrow \frac{\ln(\frac{2}{\delta}) + \ln(|\mathcal{H}|)}{2m} = \epsilon^2$$

$$\Rightarrow \epsilon = \sqrt{\frac{\ln(|\mathcal{H}|) + \ln(\frac{2}{\delta})}{2m}} \Rightarrow P\left(|E(h) - \hat{E}(h)| \leq \sqrt{\frac{\ln|\mathcal{H}| + \ln(2/\delta)}{2m}}\right) \geq 1 - \delta.$$

Finite Hypothesis Space

□ Non-separable Case

Theorem 12.1 shows that the empirical error is a good approximation of the generalization error when the sample size m is large enough. In this case $c \notin \mathcal{H}$, the better choice is to find the hypothesis with the least generalization error.

Theorem 12.1 actually points out a general learning principle:

Empirical Risk Minimization, ERM:

Let h represent the output hypothesis of the learning algorithm $\mathcal{L}$, if

$$\hat{E}(h) = \min_{h' \in \mathcal{H}} \hat{E}(h'),$$

we call the algorithm $\mathcal{L}$ satisfying the principle of empirical risk minimization.

Finite Hypothesis Space

This approach generalizes PAC learning to agnostic learning in which $c \notin \mathcal{H}$:

Definition Agnostic PAC Learnable

A hypothesis space $\mathcal{H}$ is said to be agnostic PAC learnable if there exists a learning algorithm $\mathcal{L}$ such that, for any $0 < \epsilon, \delta < 1$ and distribution $\mathcal{D}$, the learning algorithm $\mathcal{L}$ outputs a hypothesis $h \in \mathcal{H}$ satisfying

$$P(E(h) - \min_{h' \in \mathcal{H}} E(h') \leq \epsilon) \geq 1 - \delta,$$

for any $m \geq \text{poly}(1/\epsilon, 1/\delta, \text{size}(\mathbf{x}), \text{size}(c))$, where m is the number of *i.i.d.* training samples drawn from the distribution $\mathcal{D}$.

□ Theorem 12.1 show that the finite hypothesis is agnostic PAC learnable.

The sample complexity of the learning algorithm $\mathcal{L}$ is the smallest sample size m satisfying the above requirements.

Infinite Hypothesis Space

- ❑ Hypothesis spaces in real-world applications are usually **infinite**
 - All intervals in the real domain
 - All hyperplanes in the $\mathbb{R}^d$ space

- ❑ To study the learnability of such cases, we need to measure the complexity of hypothesis spaces $|\mathcal{H}|$:
 - Vapnik-Chervonenkis Dimension (VC)
 - Rademacher Complexity

VC Dimension

□ Notations

Given a hypothesis space $\mathcal{H}$ and a set of instances $D = \{x_1, x_2, \dots, x_m\}$, where each hypothesis h can label every instance in D , denoted by:

$$h|_D = \{(h(x_1), h(x_2), \dots, h(x_m))\}, \text{ with } h \in \mathcal{H}$$

- Any element of $h|_D$ is called a **dichotomy**.
- The number of dichotomies generated by the hypotheses in D over $\mathcal{H}$ increases as m increases.

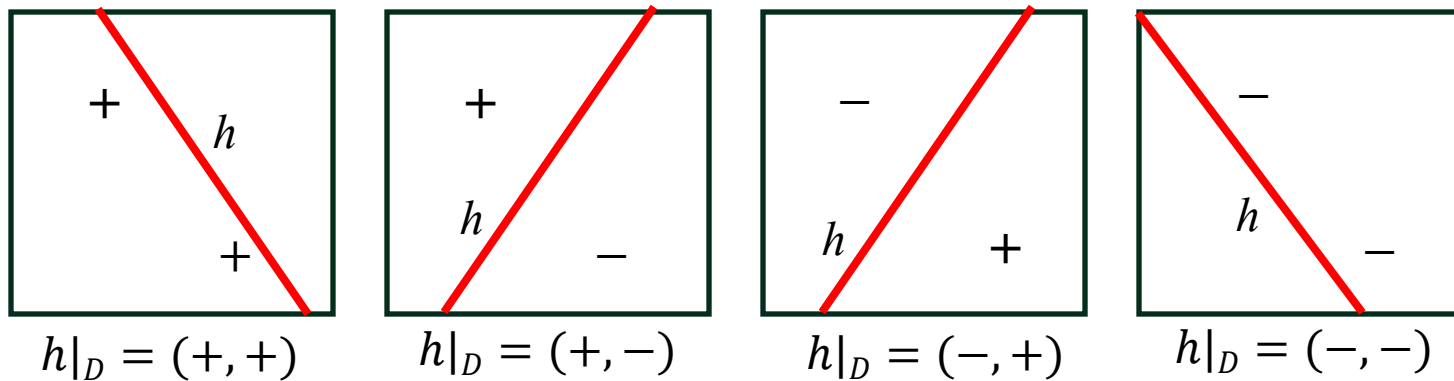
For example, in binary classification problems:

there are at most 4 dichotomies given 2 instances;

there are at most 8 dichotomies given 3 instances.

Example: in binary classification problems:

There are at most 4 dichotomies given 2 instances:



There are at most 8 dichotomies given 3 instances: 2^3

$$\begin{array}{llll} h|_D = (+, +, +) & h|_D = (+, +, -) & h|_D = (+, -, +) & h|_D = (+, -, -) \\ h|_D = (-, +, +) & h|_D = (-, +, -) & h|_D = (-, -, +) & h|_D = (-, -, -) \end{array}$$

$\nearrow$ n examples

$\searrow$ n classes

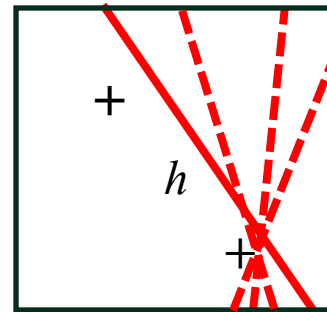
VC Dimension

□ Notations

Given a hypothesis space $\mathcal{H}$ and a set of instances $D = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$, where each hypothesis h can label every instance in D , denoted by:

$$h|_D = \{(h(\mathbf{x}_1), h(\mathbf{x}_2), \dots, h(\mathbf{x}_m))\}. \quad (\text{a dichotomy})$$

- Different hypotheses in the hypothesis space $\mathcal{H}$ may have the same or different results for the examples in D ;
- The number of dichotomies could be finite even for an infinite hypothesis space $\mathcal{H}$



$$h|_D = (+, +)$$

Example:

□ There are at most 2^m dichotomies over m instances.

□ Given $m=3$:

$$h|_D = (+, +, +) \quad h|_D = (+, +, -) \quad h|_D = (+, -, +) \quad h|_D = (+, -, -)$$

$$h|_D = (-, +, +) \quad h|_D = (-, +, -) \quad h|_D = (-, -, +) \quad h|_D = (-, -, -)$$

$$2^3 = 8$$

VC Dimension

□ Concepts

- Growth function
- Dichotomy
- Shattering

VC Dimension

Definition Growth Function

For $m \in \mathbb{N}$, the growth function $\Pi_{\mathcal{H}}(\cdot)$ of a hypothesis space $\mathcal{H}$ is defined as:

$$\Pi_{\mathcal{H}}(m) = \max_{\{x_1, x_2, \dots, x_m\} \subseteq \mathcal{X}} \text{Number of distinct dichotomies: } |\{(h(x_1), h(x_2), \dots, h(x_m)) \mid h \in \mathcal{H}\}|.$$

- The growth function gives the largest number of dichotomies that the hypothesis space $\mathcal{H}$ can generate over m instances.
- The more dichotomies, the more **representation power**, that is, the **better adaptability** to learning problems.
- The growth function describes the representation power of a hypothesis space $\mathcal{H}$, which also reflects the **complexity of the hypothesis space**.

VC Dimension

□ Dichotomy

For the binary classification problem, every possible result marked by the hypothesis h in $\mathcal{H}$ is called *dichotomy*.

□ Shattering

We say that a hypothesis space $\mathcal{H}$ can shatter a data set D if $\mathcal{H}$ can generate all possible dichotomies of D .

VC Dimension

- Remember the definition of a Probable Approximately Correct Learner (PAC) of a concept using hypothesis h learnt on a dataset whose number of examples is m :

$$P\left(|E(h) - \hat{E}(h)| \geq \epsilon\right) \leq 2\exp(-2m\epsilon^2)$$

Use a growth function $\Pi_{\mathcal{H}}(m)$ to present the relationship between the empirical error $\hat{E}(h)$ and the generalization error $E(h)$:

Theorem 12.2 [from (*)]

For any $h \in \mathcal{H}$, $m \in \mathbb{N}$, $0 < \epsilon < 1$, we have

$$P(|E(h) - \hat{E}(h)| > \epsilon) \leq 4\Pi_{\mathcal{H}}(2m) \exp\left(-\frac{m\epsilon^2}{8}\right).$$

(*): Vapnik VN, Chervonenkis A (1971) On the uniform convergence of relative frequencies of events to their probabilities. Theory Probab Its Appl 16(2):264–280

VC Dimension

Definition **Vapnik-Chervonenkis dimension**

The VC dimension of a hypothesis space $\mathcal{H}$ is the size of the largest instance set D shattered by $\mathcal{H}$

$$\text{VC}(\mathcal{H}) = \max\{m : \Pi_{\mathcal{H}}(m) = 2^m\}.$$

□ The VC dimension of $\mathcal{H}$ is d if there exists an instance set of size d shattered by $\mathcal{H}$ while there is no instance set of size $d + 1$ shattered by $\mathcal{H}$.

$$\text{VC}(\mathcal{H}) = d$$

VC Dimension

□ **Example 1:** Interval $[a, b]$ in the real domain

Let $\mathcal{H} = \{h_{[a,b]} : a, b \in \mathbb{R}, a \leq b\}$ denote the set of all closed intervals in the real domain $\mathcal{X} = \mathbb{R}$.

For every $x \in \mathcal{X}$, we have $h_{[a,b]}(x) = +1$ if $x \in [a, b]$; otherwise, $h_{[a,b]}(x) = -1$.

Letting $x_1 = 0.5, x_2 = 1.5$, then, $\{x_1, x_2\}$ is shattered by the hypotheses $\{h_{[0,1]}, h_{[0,2]}, h_{[1,2]}, h_{[2,3]}\}$ from $\mathcal{H}$.

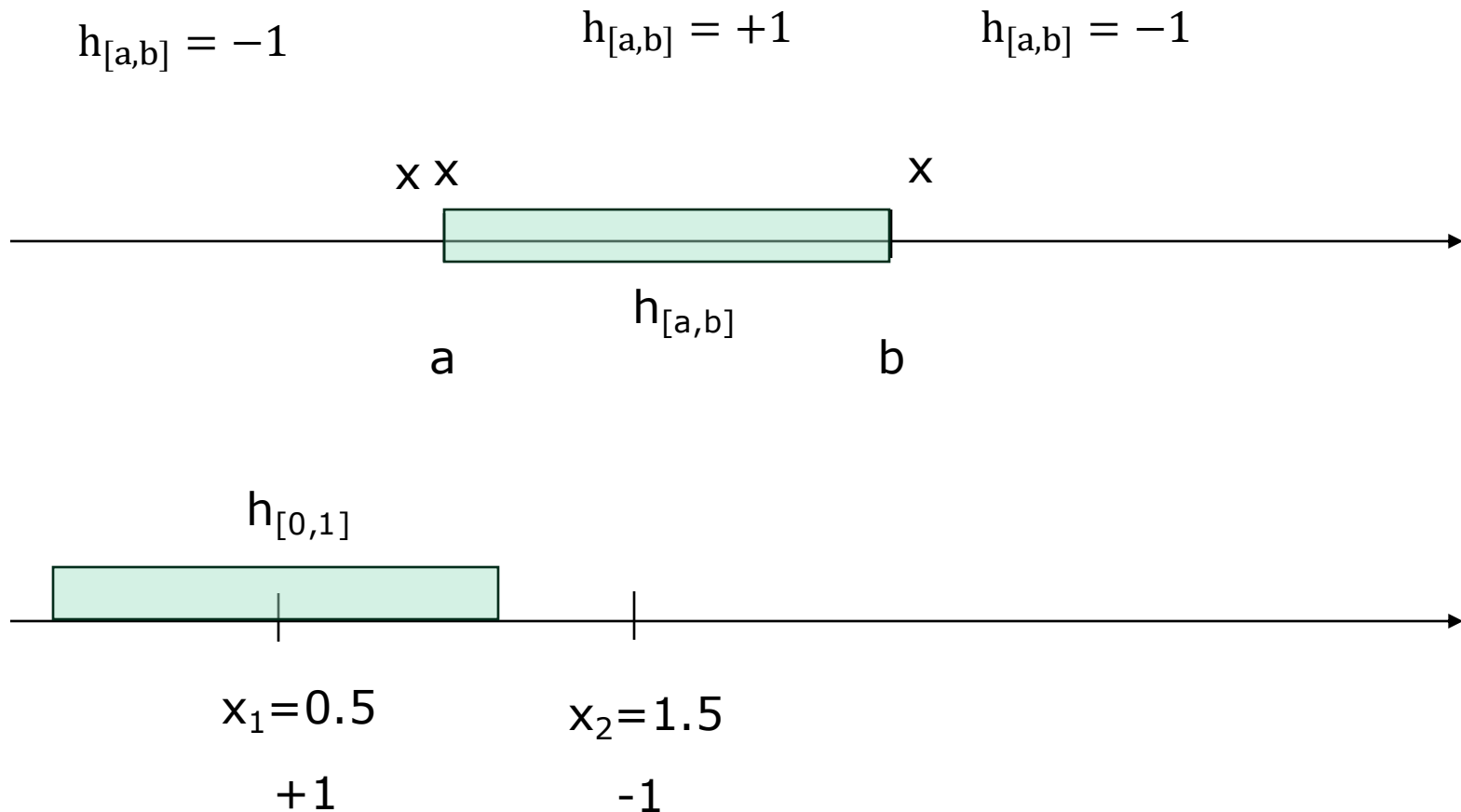
Hence VC dimension of $\mathcal{H}$ is at least 2.

However, there is no hypothesis $h_{[a,b]} \in \mathcal{H}$ that can generate the dichotomy $\{(x_3, +), (x_4, -), (x_5, +)\}$ for a data set containing any 3 instances $\{x_3, x_4, x_5\}$, where $x_3 < x_4 < x_5$,

Hence, the VC dimension of $\mathcal{H}$ is 2.

Example on Interval $[a,b]$ in the real domain

$h_{[a,b]}$



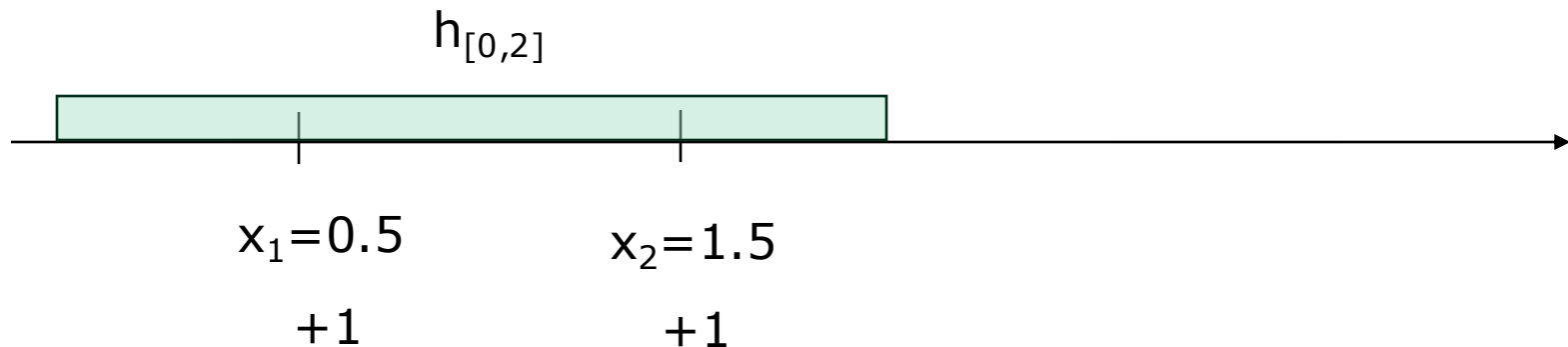
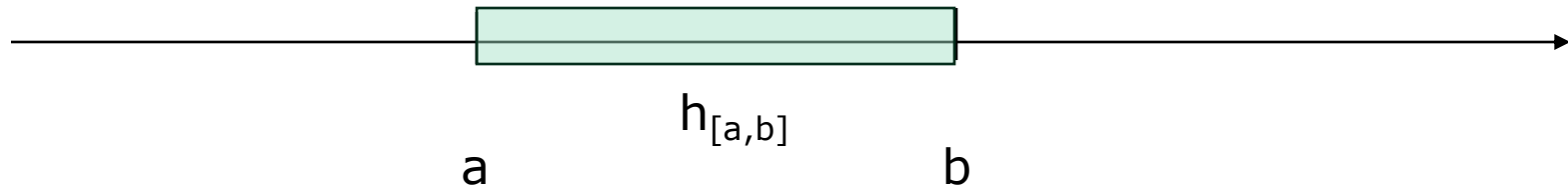
Example on Interval $[a,b]$ in the real domain

$h_{[a,b]}$

$$h_{[a,b]} = -1$$

$$h_{[a,b]} = +1$$

$$h_{[a,b]} = -1$$



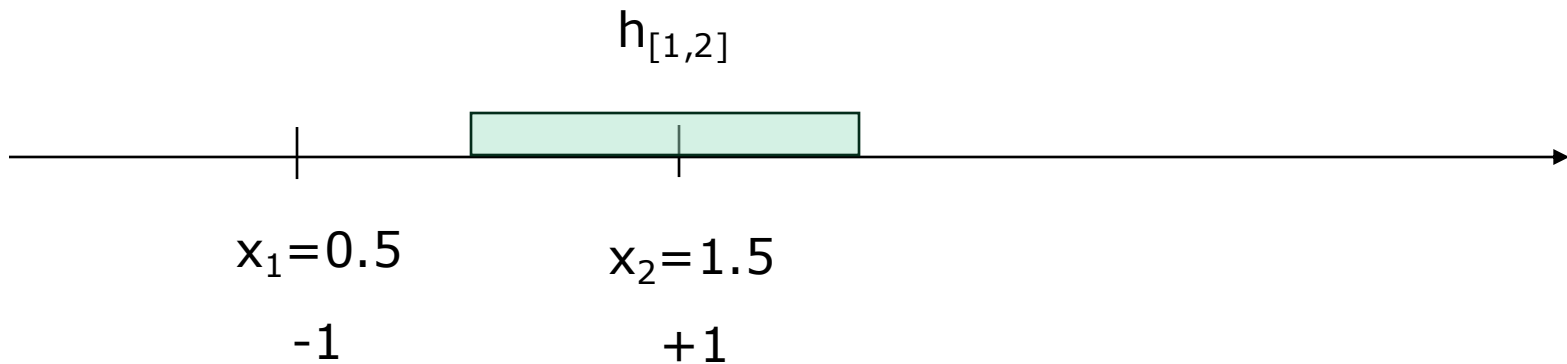
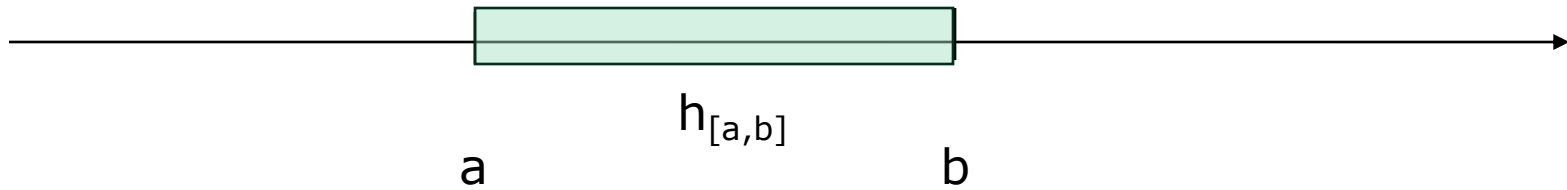
Example on Interval $[a,b]$ in the real domain

$h_{[a,b]}$

$$h_{[a,b]} = -1$$

$$h_{[a,b]} = +1$$

$$h_{[a,b]} = -1$$



Example on Interval $[a,b]$ in the real domain

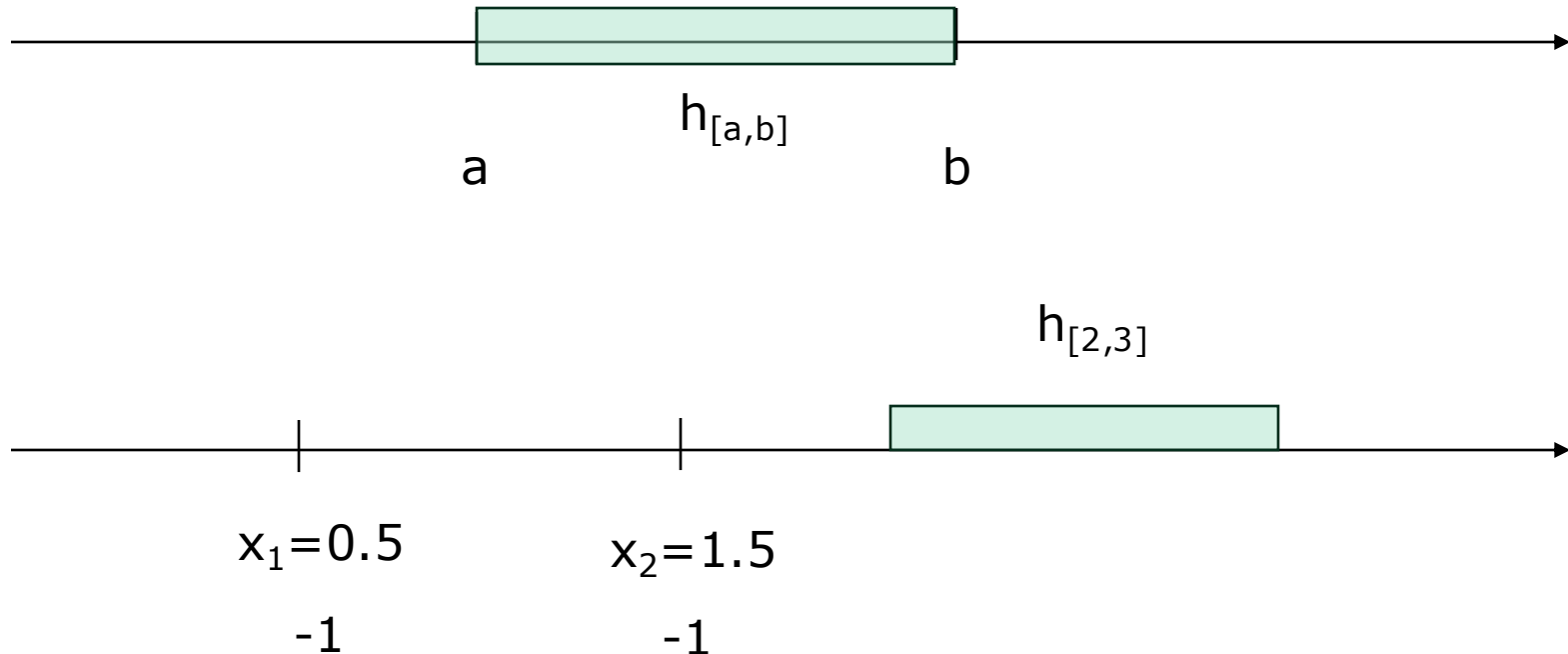
$h_{[a,b]}$

x_1 and x_2 are shattered by the hypothesis space $\mathcal{H} = \{h_{[0,1]}, h_{[0,2]}, h_{[1,2]}, h_{[2,3]}\}$ because all together produce all the possible labellings (dichotomies)

$$h_{[a,b]} = -1$$

$$h_{[a,b]} = +1$$

$$h_{[a,b]} = -1$$



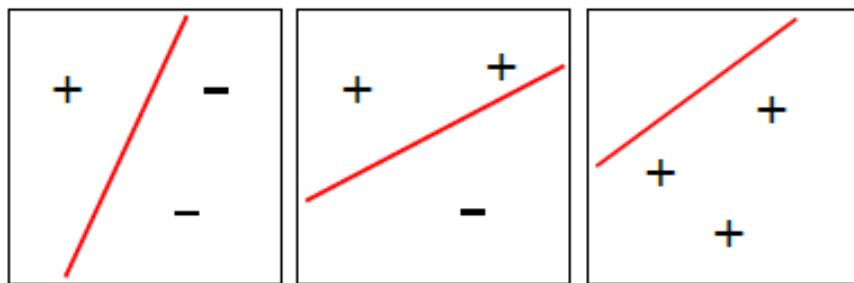
VC Dimension: Example 2

□ Linear separators in the 2-D real plane

Let $\mathcal{H}$ denote the set of all linear separators in the 2-dimensional real plane $\mathcal{X} = \mathbb{R}^2$.

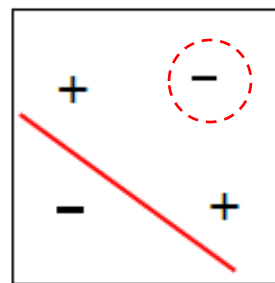
From Figure we see that there exists a data set of size 3 shattered by $\mathcal{H}$, whereas there is no instance set of size 4 shattered by $\mathcal{H}$. Hence, the VC dimension of the hypothesis space $\mathcal{H}$ of all linear separators in the 2-dimensional real plane is

3.



All of the $2^3=8$ dichotomies can be made by linear separators

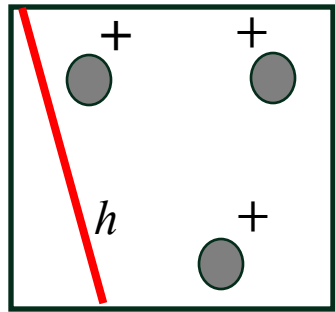
(a) 3 instances.



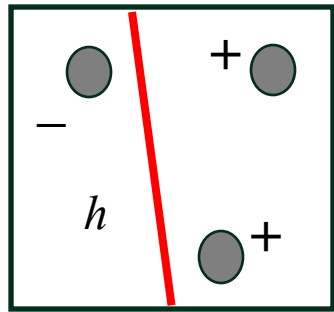
At least one of the $2^4=16$ dichotomies cannot be made by linear separators

(b) 4 instances.

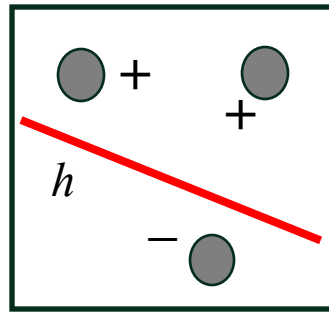
D of size 3 is shattered by linear separators



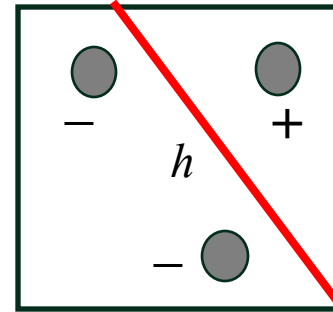
$$h|_D = (+, +, +)$$



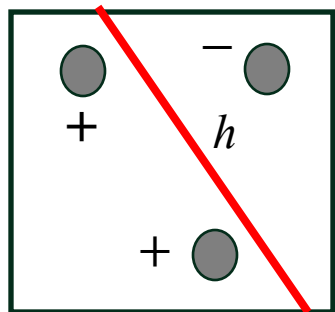
$$h|_D = (-, +, +)$$



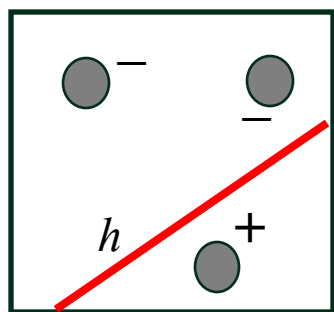
$$h|_D = (+, -, +)$$



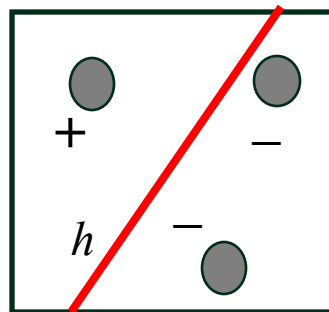
$$h|_D = (-, -, +)$$



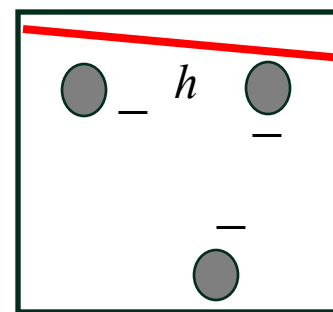
$$h|_D = (+, +, -)$$



$$h|_D = (-, +, -)$$



$$h|_D = (+, -, -)$$



$$h|_D = (-, -, -)$$

VC Dimension

- The relationship between the VC dimension and the growth function $\Pi_{\mathcal{H}}(m)$:
- growth function $\Pi_{\mathcal{H}}(m)$:
maximum number of dichotomies that $\mathcal{H}$ can generate from a data set of size m
- maximum size m of the data set shattered by $\mathcal{H}$

Sauer's Lemma

If the VC dimension of a hypothesis space $\mathcal{H}$ is d , then, for any $m \in \mathbb{N}$, we have

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i}.$$

Example:

□ Given $d=m=3$:

$$h|_D = (+, +, +) \quad h|_D = (+, +, -) \quad h|_D = (+, -, +) \quad h|_D = (+, -, -)$$

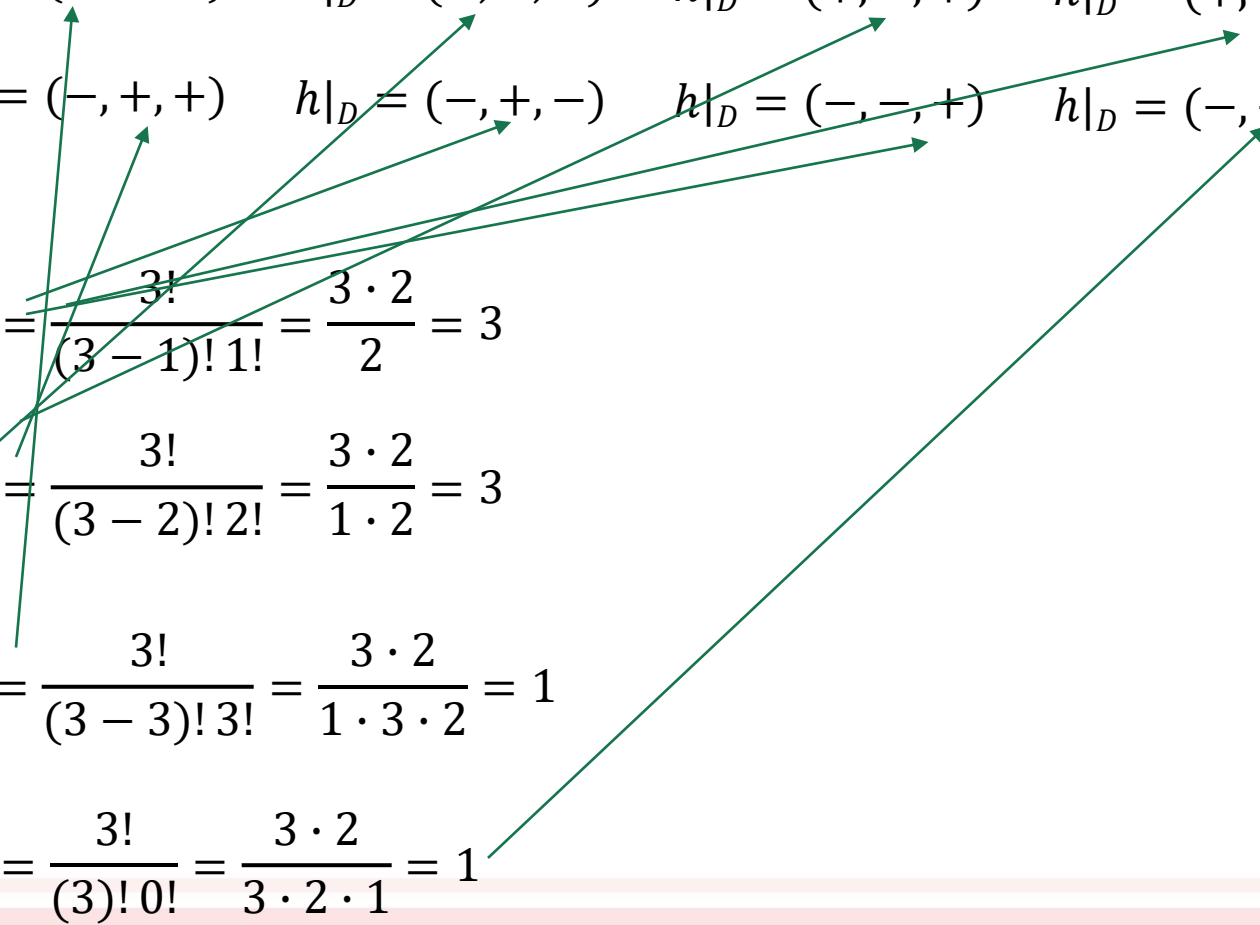
$$h|_D = (-, +, +) \quad h|_D = (-, +, -) \quad h|_D = (-, -, +) \quad h|_D = (-, -, -)$$

$$\binom{3}{1} = \frac{3!}{(3-1)!1!} = \frac{3 \cdot 2}{2} = 3$$

$$\binom{3}{2} = \frac{3!}{(3-2)!2!} = \frac{3 \cdot 2}{1 \cdot 2} = 3$$

$$\binom{3}{3} = \frac{3!}{(3-3)!3!} = \frac{3 \cdot 2}{1 \cdot 3 \cdot 2} = 1$$

$$\binom{3}{0} = \frac{3!}{(3)!0!} = \frac{3 \cdot 2}{3 \cdot 2 \cdot 1} = 1$$



VC Dimension: upper bound of $\Pi_{\mathcal{H}}(m)$

We can calculate the upper bound of the growth function:

Corollary:

If the VC dimension of a hypothesis space $\mathcal{H}$ is d , then, for any integer $m \geq d$, we have

$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{e \cdot m}{d} \right)^d.$$

where e is the Euler constant $(\lim_{n \rightarrow \infty} (1 + 1/n)^n)$

VC Dimension

We have the VC bound:

Theorem 12.3

If the VC dimension of a hypothesis $\mathcal{H}$ is d , then, for any $m > d, 0 < \delta < 1$, and $h \in \mathcal{H}$, we have

$$P \left(|E(h) - \hat{E}(h)| \leq \sqrt{\frac{8d \ln \frac{2em}{d} + 8 \ln \frac{4}{\delta}}{m}} \right) \geq 1 - \delta.$$

Proof:

from Theorem 12.2: $P(|E(h) - \hat{E}(h)| > \epsilon) \leq 4 \Pi_{\mathcal{H}}(2m) \exp(-\frac{m\epsilon^2}{8})$.

From the corollary $\Pi_{\mathcal{H}}(m) \leq \left(\frac{e \cdot m}{d}\right)^d$ and substituting above, we have:

$$4 \Pi_{\mathcal{H}}(2m) \exp(-\frac{m\epsilon^2}{8}) \leq 4 \left(\frac{2em}{d}\right)^d \exp(-\frac{m\epsilon^2}{8}) = \delta$$

VC Dimension

□ Resolving for ϵ gives:

$$\epsilon = \sqrt{\frac{8d \ln \frac{2em}{d} + 8 \ln \frac{4}{\delta}}{m}},$$

□ Knowing that

$$P(|E(h) - \hat{E}(h)| > \epsilon) \leq \delta$$

$$\Rightarrow P(|E(h) - \hat{E}(h)| \leq \epsilon) \geq 1 - \delta$$

□ Substituting ϵ in the above expression we have the conclusion.

VC Dimension

From the VC bound of Theorem 12.3:

$$P \left(|E(h) - \hat{E}(h)| \leq \sqrt{\frac{8d \ln \frac{2em}{d} + 8 \ln \frac{4}{\delta}}{m}} \right) \geq 1 - \delta.$$

- We observe that the generalization error bound is dependent only on the sample size m and converges at a rate of $O\left(\frac{1}{\sqrt{m}}\right)$.
- VC bound is independent of the data distribution $\mathcal{D}$ and the data set D .

It is **distribution-free** and **data-independent**

VC Dimension

Similar to theorem 12.1, we see from theorem 12.3 that when the VC dimension of the space is limited and the sample size m is large enough, **the empirical error $\hat{E}(h)$ is a good approximation of the generalization error $E(h)$** .

Therefore, for the learning algorithm $\mathcal{L}$ satisfying the principle of empirical risk minimization, there is the following theorem:

Theorem 12.4

Every hypothesis space $\mathcal{H}$ with a finite VC dimension is (agnostic) PAC learnable.

Rademacher Complexity

- ❑ The VC bound is **distribution-free** and **data-independent**, which makes the analysis of generalization error bound “universal”.
- ❑ Since it does not take the data set into account, the VC bound is generally *loose*.
- ❑ Can the data distribution be taken into account?

Rademacher complexity is another characterization of the complexity of the hypothesis space, and the difference from the VC dimension lies in **consideration of data distribution in some sense**.

Rademacher Complexity

Given a data set $D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$,
the empirical error of a hypothesis h is given by

$$\begin{aligned}\hat{E}(h) &= \frac{1}{m} \sum_{i=1}^m \mathbb{I}(h(\mathbf{x}_i) \neq y_i) \\ &= \frac{1}{m} \sum_{i=1}^m \frac{1 - y_i h(\mathbf{x}_i)}{2} \\ &= \frac{1}{2} - \frac{1}{2m} \sum_{i=1}^m y_i h(\mathbf{x}_i)\end{aligned}$$

Assuming y_i and $h(\mathbf{x}_i)$ giving either +1 or -1.

If they agree on $\mathbf{x}_i$ the term $\frac{1 - y_i h(\mathbf{x}_i)}{2}$ gives $\frac{1 - 1 \cdot 1}{2} = 0$ or $\frac{1 - (-1 \cdot -1)}{2} = 0$

If they do not agree, its term $\frac{1 - y_i h(\mathbf{x}_i)}{2}$ accounts for 1 error.

Rademacher Complexity

Given a data set $D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$, the empirical error of a hypothesis h is given by

$$\hat{E}(h) = \frac{1}{2} - \frac{1}{2m} \sum_{i=1}^m y_i h(\mathbf{x}_i)$$

- where $\frac{1}{m} \sum_{i=1}^m y_i h(\mathbf{x}_i)$ represents the consistency between the predicted values $h(\mathbf{x}_i)$ and the ground-truth labels y_i .
- It takes the maximum value 1 if $h(\mathbf{x}_i) = y_i$, for all $i \in \{1, 2, \dots, m\}$,
- The hypothesis with the smallest empirical error is

$$\arg \max_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m y_i h(\mathbf{x}_i).$$

Rademacher Complexity

- The hypothesis with the smallest empirical error is

$$\arg \max_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m y_i h(\mathbf{x}_i).$$

- The data set may have been corrupted by some noises.

In such cases, sometimes it is better to select a hypothesis $\mathcal{H}$ that has considered the influence of random noises

$$\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(\mathbf{x}_i).$$

It is likely that we cannot find the maximum value since $\mathcal{H}$ is infinite, so we use supremum.

- The Rademacher random variable: σ_i

takes value +1 or -1 with an equal probability of 0.5.

(*) sup: The **supremum** (abbreviated **sup**) of a subset S of a partially ordered set P that is greater than or equal to each element of S.

Rademacher Complexity

□ Consider all hypothesis in $\mathcal{H}$ and take the expectation as

$$\mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(\mathbf{x}_i) \right] .$$

- where $\sigma = \{\sigma_1, \sigma_2, \dots, \sigma_m\}$.
- Takes value in $[0, 1]$, expresses the representation power of the hypothesis $\mathcal{H}$.
 - Equals to 0 when $|\mathcal{H}| = 1$, that is, there is only one hypothesis in $\mathcal{H}$;
 - $\mathbb{E}_{\sigma}[h(\mathbf{x}_i)] = 1 \cdot 0.5 \cdot h(\mathbf{x}_i) - 1 \cdot 0.5 \cdot h(\mathbf{x}_i) = 0$

Rademacher Complexity

□ Consider all hypothesis in $\mathcal{H}$ and take the expectation as

$$\mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(\mathbf{x}_i) \right] .$$

- where $\sigma = \{\sigma_1, \sigma_2, \dots, \sigma_m\}$.
- Takes value in $[0, 1]$, expresses the representation power of the hypothesis $\mathcal{H}$.
 - Equals to 1 when $|\mathcal{H}| = 2^m$ and $\mathcal{H}$ shatters D , that is, for any σ , there exists a hypothesis such that $h(\mathbf{x}_i) = \sigma_i (i = 1, 2, \dots, m)$ (perfectly matches the observed random noise).

Rademacher Complexity

Definition Empirical Rademacher complexity

The empirical Rademacher complexity of a function space $\mathcal{F}$ with respect to Z is defined as

$$\hat{R}_Z(\mathcal{F}) = \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^m \sigma_i f(z_i) \right].$$

- Let $\mathcal{F} : \mathcal{Z} \rightarrow \mathbb{R}$ be a real-valued function space, and $Z = \{z_1, z_2, \dots, z_m\}$ be a set of *i.i.d.* instance, where $z_i \in \mathcal{Z}$.
- The empirical Rademacher complexity **measures the correlation between the function space $\mathcal{F}$ and the random noise** in the data set Z .

Rademacher Complexity

Definition **Rademacher complexity**

The Rademacher complexity of a function space $\mathcal{F}$ with respect to a distribution $\mathcal{D}$ over Z is defined as

$$R_m(\mathcal{F}) = \mathbb{E}_{Z \subseteq \mathcal{Z}: |Z|=m} \left[\hat{R}_Z(\mathcal{F}) \right].$$

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- ▣ Based on the Rademacher complexity, we can define the generalization error bound of function space $\mathcal{F}$.

Rademacher Complexity

Theorem 12.5

Let $\mathcal{F} : \mathcal{Z} \rightarrow [0, 1]$ be a real-valued function space, and $Z = \{z_1, z_2, \dots, z_m\}$ be a set of *i.i.d.* samples drawn from $\mathcal{D}$ over $\mathcal{Z}$.

Then, for any $\delta \in (0, 1)$ and $f \in \mathcal{F}$, the following holds with a probability of at least $1 - \delta$:

$$\mathbb{E}[f(z)] \leq \frac{1}{m} \sum_{i=1}^m f(z_i) + 2R_m(\mathcal{F}) + \sqrt{\frac{\ln(1/\delta)}{2m}},$$
$$\mathbb{E}[f(z)] \leq \frac{1}{m} \sum_{i=1}^m f(z_i) + 2\hat{R}_Z(\mathcal{F}) + 3\sqrt{\frac{\ln(2/\delta)}{2m}}.$$

- Since $\mathcal{F}$ in Theorem 12.5 is a real-valued function over the interval $[0, 1]$, Theorem 12.5 is applicable **to regression problems** only

Rademacher Complexity

Theorem 12.6 : Generalization error bound

Let $\mathcal{H} : \mathcal{X} \rightarrow \{-1, +1\}$, be a hypothesis space and $D = \{x_1, x_2, \dots, x_m\}$ be a set of *i.i.d.* instances drawn from $\mathcal{D}$ over $\mathcal{X}$. Then, for any $\delta \in (0, 1)$ and $h \in \mathcal{H}$, the following holds with a probability of at least $1 - \delta$:

$$E(h) \leq \hat{E}(h) + R_m(\mathcal{H}) + \sqrt{\frac{\ln(1/\delta)}{2m}},$$

$$E(h) \leq \hat{E}(h) + \hat{R}_D(\mathcal{H}) + 3\sqrt{\frac{\ln(2/\delta)}{2m}}.$$

- Theorem 12.6 fits binary classification problems.

Rademacher Complexity

Theorem 12.3 If the VC dimension of a hypothesis $\mathcal{H}$ is d , then, for any $m > d$, $\delta \in (0, 1)$, and $h \in \mathcal{H}$, we have

$$P \left(\left| E(h) - \hat{E}(h) \right| \leq \sqrt{\frac{8d \ln \frac{2em}{d} + 8 \ln \frac{4}{\delta}}{m}} \right) \geq 1 - \delta. \quad (12.29)$$

Theorem 12.6 Let $\mathcal{H} : \mathcal{X} \rightarrow \{-1, +1\}$ be a hypothesis space and $D = \{x_1, x_2, \dots, x_m\}$ be a set of i.i.d. instances drawn from $\mathcal{D}$ over $\mathcal{X}$. Then, for any $\delta \in (0, 1)$ and $h \in \mathcal{H}$, the following holds with a probability of at least $1 - \delta$:

$$E(h) \leq \hat{E}(h) + R_m(\mathcal{H}) + \sqrt{\frac{\ln(1/\delta)}{2m}}, \quad (12.47)$$

$$E(h) \leq \hat{E}(h) + \hat{R}_D(\mathcal{H}) + 3\sqrt{\frac{\ln(2/\delta)}{2m}}. \quad (12.48)$$

- Theorem 12.3(VC bound) is *distribution-free* and *data-independent*;
- Theorem 12.6(Rademacher bound) is related to $\mathcal{D}$ and D .

The generalization error bound based on Rademacher complexity depends on the data distribution of the specific learning problem, similar to "tailor-made" for the problem. Therefore, it is usually tighter than the generalization error bound based on VC dimension.

Rademacher Complexity

- For the Rademacher complexity and the growth function:

Theorem 12.7

The Rademacher complexity $R_m(\mathcal{H})$ and the growth function $\Pi_{\mathcal{H}}(m)$ of a hypothesis space $\mathcal{H}$ satisfy :

$$R_m(\mathcal{H}) \leq \sqrt{\frac{2 \ln \Pi_{\mathcal{H}}(m)}{m}}.$$

- From Theorem 12.6、Theorem 12.7、Corollary 12.2:

$$E(h) \leq \hat{E}(h) + \sqrt{\frac{2d \ln \frac{em}{d}}{m}} + \sqrt{\frac{\ln(1/\delta)}{2m}}.$$

- In other words, we can derive the VC bound from the Rademacher complexity and the growth function.

Stability

- Whether the generalization performance is analyzed based on VC dimension and Rademacher complexity, the results are independent of the specific learning algorithm, which enables people to consider the nature of the learning problem itself without the specific learning algorithm.
- But on the other hand, in order to obtain the analysis results related to the algorithm, we need to find another way.
- Stability analysis is a direction worthy of attention in this field.
 - Whether the output of the algorithm changes greatly when the input (training set) changes is investigated.

Stability

□ Two changes of training set

Given $D = \{z_1 = (\mathbf{x}_1, y_1), z_2 = (\mathbf{x}_2, y_2), \dots, z_m = (\mathbf{x}_m, y_m)\}$, $\mathbf{x}_i \in \mathcal{X}$ are *i.i.d.* instances drawn from distribution $\mathcal{D}$ and $y_i \in \{-1, +1\}$. Let $\mathcal{H} : \mathcal{X} \rightarrow \{-1, +1\}$ be a hypothesis space, and $\mathcal{L}_D \in \mathcal{H}$ be the hypothesis learned by a learning algorithm $\mathcal{L}$ on the training set D

- Let $D^{\setminus i}$ denote the set D with the i th sample excluded, that is,

$$D^{\setminus i} = \{z_1, z_2, \dots, z_{i-1}, z_{i+1}, \dots, z_m\},$$

- Let D^i denote the set D with the i th sample replaced, that is,

$$D^i = \{z_1, z_2, \dots, z_{i-1}, z'_i, z_{i+1}, \dots, z_m\},$$

where $z'_i = (\mathbf{x}'_i, y'_i)$, $\mathbf{x}'_i$ follow distribution $\mathcal{D}$ and is dependent of D .

Stability

□ Loss function

$\ell(\mathcal{L}_D(\mathbf{x}), y) : \mathcal{Y} \times \mathcal{Y} \mapsto \mathbb{R}^+$ abbreviated as $\ell(\mathcal{L}_D, \mathbf{z})$, characterizes the difference between the predicted label $\mathcal{L}_D(\mathbf{x})$ and the ground-truth label y

● Generalization loss $\ell(\mathcal{L}, D) = \mathbb{E}_{\mathbf{x} \in \mathcal{X}, \mathbf{z} = (\mathbf{x}, y)} [\ell(\mathcal{L}_D, \mathbf{z})]$.

● Empirical loss $\hat{\ell}(\mathcal{L}, D) = \frac{1}{m} \sum_{i=1}^m \ell(\mathcal{L}_D, \mathbf{z}_i)$.

● Leave-one-out loss $\ell_{loo}(\mathcal{L}, D) = \frac{1}{m} \sum_{i=1}^m \ell(\mathcal{L}_{D \setminus i}, \mathbf{z}_i)$.

Stability

Definition **Uniform stability**

For any $x \in \mathcal{X}$, $z = (x, y)$, a learning algorithm $\mathcal{L}$ satisfies

$$|\ell(\mathcal{L}_D, z) - \ell(\mathcal{L}_{D \setminus i}, z)| \leq \beta, i = 1, 2, \dots, m,$$

$\mathcal{L}$ is said to satisfy the β —uniform stability with respect to loss function ℓ

- If a learning algorithm $\mathcal{L}$ satisfies the β —uniform stability with respect to a loss function ℓ , then

$$\begin{aligned} & |\ell(\mathcal{L}_D, z) - \ell(\mathcal{L}_{D^i}, z)| \\ & \leq |\ell(\mathcal{L}_D, z) - \ell(\mathcal{L}_{D \setminus i}, z)| + |\ell(\mathcal{L}_{D^i}, z) - \ell(\mathcal{L}_{D \setminus i}, z)| \\ & \leq 2\beta \end{aligned}$$

which means that the stability of excluding an instance implies the stability of replacing an instance.

Stability

□ If the loss function ℓ is bounded

For all D and $z = (x, y)$, $0 \leq \ell(\mathcal{L}_D, z) \leq M$, then

Theorem 12.8 Given a data set D with m *i.i.d.* instances drawn from the distribution $\mathcal{D}$. If a learning algorithm $\mathcal{L}$ satisfies the β –uniform stability with respect to a loss function ℓ upper bounded by M , then, for any $m \geq 1$, and $0 < \delta < 1$, the following holds with a probability of at least $1 - \delta$:

$$\ell(\mathcal{L}, \mathcal{D}) \leq \hat{\ell}(\mathcal{L}, D) + 2\beta + (4m\beta + M) \sqrt{\frac{\ln(1/\delta)}{2m}},$$

$$\ell(\mathcal{L}, \mathcal{D}) \leq \ell_{loo}(\mathcal{L}, D) + \beta + (4m\beta + M) \sqrt{\frac{\ln(1/\delta)}{2m}}.$$

Stability

- Theorem 12.8 shows the generalization error bound of the learning algorithm $\mathcal{L}$ derived from the stability analysis.
 - The convergence rate between the empirical error and the generalization error is $\beta\sqrt{m}$;
 - When $\beta = O(\frac{1}{m})$, the convergence rate becomes $O(\frac{1}{\sqrt{m}})$.
 - which is consistent with those of VC bound and Rademacher bound.

Stability

- The stability analysis of learning algorithm focuses on

$$|\hat{\ell}(\mathcal{L}, D) - \ell(\mathcal{L}, \mathcal{D})|;$$

- the complexity analysis of the hypothesis space considers

$$\sup_{h \in \mathcal{H}} |\hat{E}(h) - E(h)|.$$

In other words, the stability analysis does not necessarily consider every hypothesis, but only analyzes the generalization error bound of the output hypothesis $\mathcal{L}_D$

What is the relationship between stability and learnability?

Stability

First of all, it must be assumed that $\beta\sqrt{m} \rightarrow 0$, only in this way can the stable learning algorithm have a certain generalization ability, that is, the experience loss converges to the generalization loss, otherwise the learnability is impossible. For the learning algorithm satisfying ERM principle, there are the following theorems:

Theorem **12.9**

If an ERM learning algorithm $\mathcal{L}$ is stable, then the hypothesis space $\mathcal{H}$ is learnable.

- The stability of the learning algorithm leads to the learnability of the hypothesis space
- The stability and assumption space can be connected by loss function ℓ

Conclusion

□ Overview

- Concerns
- Some concepts and notations

□ Learnability

- What is learning
- What is learnable
- The impact of the complexity of hypothesis spaces on learnability
 - Finite hypothesis space
 - Infinite hypothesis space: VC dimension
 - Infinite hypothesis space: Rademacher complexity

□ Stability

Thanks!
